Prob 3. (2 PT. Ch 7. 1.2 a, b)

(a) Show that $u_1 = xy$ & $u_2 = x^2 - y^2$ are harmonic in $\mathbb{R}^2$.

Sol-n:

Based on the def-n of harmonic f-ns (given in previous Pr. 1);

① $u_1(x,y) = x \cdot y$ & $u_2(x,y) = x^2 - y^2$ are both poly-s of 2 variables, hence both, u_1, u_2, are from $C^\infty(\mathbb{R}^2)$

② $$\left.\begin{array}{ll} \frac{\partial u_1}{\partial x} = y & \frac{\partial u_1}{\partial y} = x \\ \frac{\partial^2 u_1}{\partial x^2} = 0 & \frac{\partial^2 u_1}{\partial y^2} = 0 \end{array}\right\} \Rightarrow \frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0 + 0 \equiv 0 \checkmark$$

satisfies the Laplace eq-n, hence harmonic.

Similarly

$$\left.\begin{array}{ll} \frac{\partial u_2}{\partial x} = 2x & \frac{\partial u_2}{\partial y} = -2y \\ \frac{\partial^2 u_2}{\partial x^2} = 2 & \frac{\partial^2 u_2}{\partial y^2} = -2 \end{array}\right\} = \frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 2 + (-2) \equiv 0 \checkmark$$

then u_2 is also satisfying the Laplace eq-n, making both f-ns harmonic. ∎

(b) Find all <u>homog-s</u> poly-s of <u>degree 2</u> that are <u>harmonic</u> in $\mathbb{R}^2$.

Sol-n:

(1) $\boxed{u(x,y) = ax^2 + bxy - ay^2}$ (?4) all

Note the f-n $u(x,y)$ in (1), first, $u \in C^\infty(\mathbb{R}^2) \subset C^1(\mathbb{R}^2)$

(i) $u(tx, ty) = a(tx)^2 + b(tx)\cdot(ty) - a(ty)^2 = t^2(ax^2 + bxy - ay^2) =$

$= t^2 \cdot u(x,y) \Rightarrow$ homog-s poly-n of degree 2, $n = 2$.

(ii) $u_{xx} = 2a$ & $u_{yy} = -2a \Rightarrow \Delta u = 0$ &

$u(x,y) \in C^\infty(\mathbb{R}^2)$ hence a harmonic. ∎